# CivicPulse: An AI-Powered Government Services Navigator

**A Technical Report**

An open-source, multilingual, RAG-powered agentic system for navigating complex government licensing and permitting requirements.

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| **Course** | MSIS 522: AI-Powered Products |
| **Institution** | University of Washington, Michael G. Foster School of Business |
| **Date** | March 2026 |

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|---|---|
| **Live Application** | civic-pu1se.netlify.app |
| **Code Repository** | github.com/Meerxn/civic-pulse |
| **Presentation Deck** | Gamma Slide Deck |

**Abstract.** This report presents a comprehensive technical overview of CivicPulse, an AI-powered government services navigator built for the City of Seattle. The system helps entrepreneurs particularly those in non-English-speaking communities obtain personalized, step-by-step roadmaps for business licensing

and permitting. It does so through a multi-agent architecture orchestrated by LangGraph, a Retrieval Augmented Generation (RAG) pipeline grounded in official government documents, and a multilingual conversational interface. The report covers the system's design rationale, architecture, technology stack, key features, performance evaluation, known limitations, security and ethical considerations, and future development trajectory.

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# 1. Executive Summary

Starting a new business in Seattle requires an entrepreneur to navigate a fragmented web of city, county, and state regulations. The process involves multiple government agencies, each with its own forms, deadlines, fees, and terminology. This bureaucratic friction creates significant barriers to economic participation, particularly for first-time founders and non-English-speaking communities.

CivicPulse was built to address this gap. It is an AI-powered government services navigator that provides personalized, step-by-step roadmaps for business licensing and permitting. A user types a question in plain language in any of six supported languages and receives a structured action plan grounded entirely in official government documents. The system is powered by a multi-agent pipeline orchestrated through LangGraph, with specialist agents for business permits and health regulations, a synthesizer agent that assembles a coherent roadmap, and a Retrieval-Augmented Generation (RAG) pipeline backed by a FAISS vector index over 1,184 document chunks scraped from nine government websites.

This report details the technical architecture, implementation decisions, and design rationale behind CivicPulse. It also addresses the system's performance metrics, current limitations, security and responsible AI measures, and a roadmap for future development.

# 2. Problem Domain

## 2.1 The Fragmentation of Government Services

The core problem CivicPulse addresses is the severe fragmentation of government

services required to start a business in Seattle. An aspiring entrepreneur must interact with at least three distinct layers of government, each operating independently with its own websites, application processes, and requirements.

The **City of Seattle** manages local business licenses, zoning regulations, and specific health codes. **King County** oversees county-wide public health permits and food handler requirements. **Washington State** handles foundational business registration, tax filings, and corporate entity formation. There is no single, unified guide that consolidates these disparate requirements into a coherent, end-to-end process. Information is scattered across dozens of websites, written in dense bureaucratic language, and lacks a clear starting point.

This complexity creates a significant regulatory burden. A 2024 survey by the U.S. Chamber of Commerce found that 51% of small businesses report that licensing and permit requirements make it harder for them to grow. The result is a frustrating cycle of research, guesswork, and costly errors. This administrative friction is a contributing factor to the high failure rate of new enterprises; according to the U.S. Bureau of Labor Statistics, 20.4% of new businesses fail within their first year.

## 2.2 Who Is Most Affected

While this complexity affects every entrepreneur, it creates near-insurmountable barriers for specific communities. Over 21% of Seattle residents speak a language other than English at home. For these residents, the process means independently navigating multiple English-only government websites. First-time entrepreneurs and low-income residents, who often lack the experiential context or financial resources to hire consultants, face similar challenges.

# 3. Solution Overview

## 3.1 Core Value Proposition

CivicPulse transforms the fragmented and confusing process of starting a business into a simple, guided, conversational experience. A user asks a question in plain language and receives a personalized, step-by-step roadmap that details the specific forms, fees, agencies, and deadlines involved. Every step is linked to its official government source, and the entire interaction can take place in any of six supported languages.

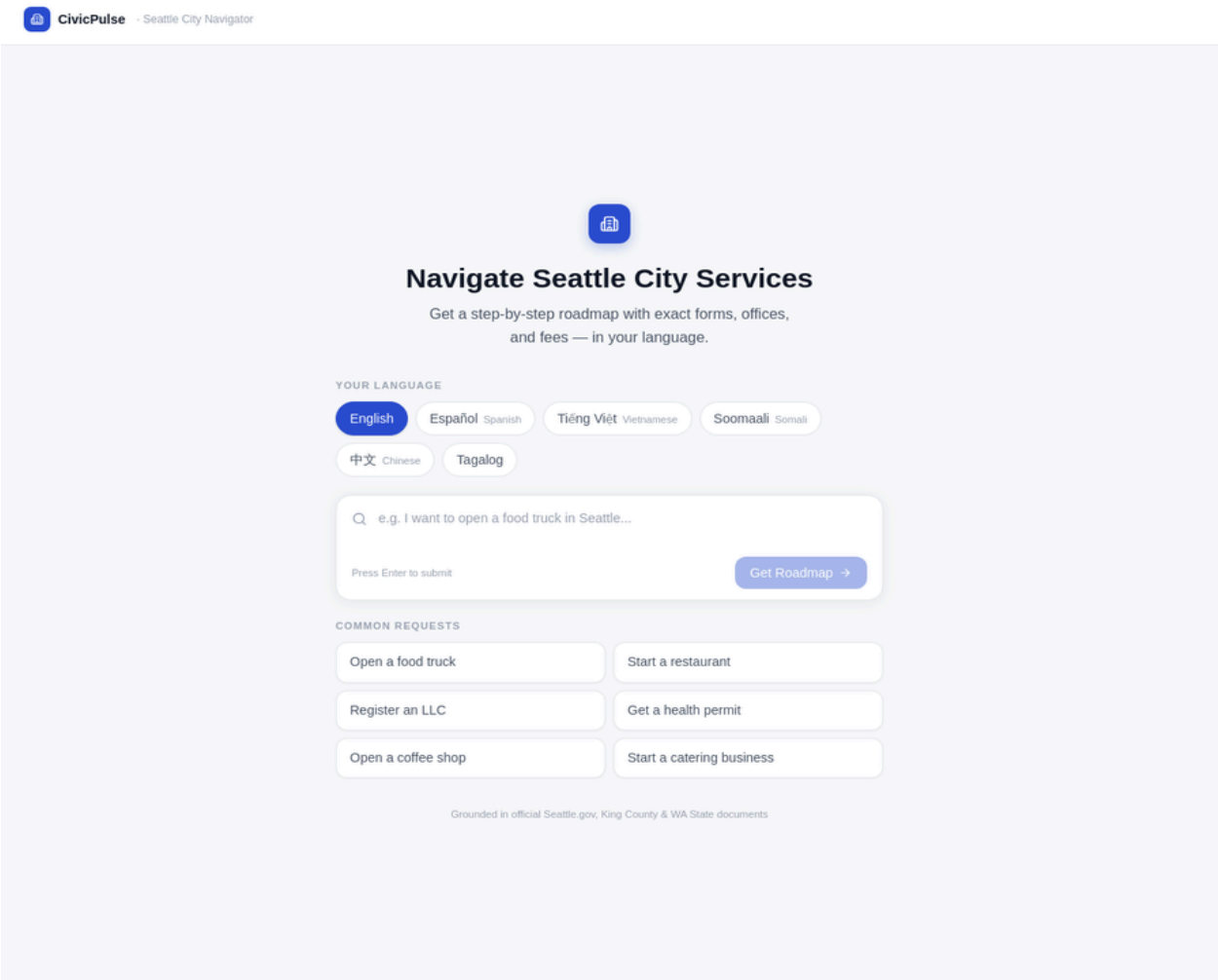

Figure 1: The CivicPulse landing page, showing the multilingual interface and common request buttons.

## 3.2 Key Features

| Feature | Description |
|---|---|
| **Smart Routing** | The system analyzes the user's query to determine whether it involves business permits, health permits, or both, and routes the request to the appropriate specialist AI agents. |
| **Multilingual Support** | CivicPulse auto-detects the user's language and responds in kind. Six languages are supported: English, Spanish, Vietnamese, Somali, Chinese, and Tagalog. |
| **Grounded Generation** | All information is derived from a curated knowledge base of official government websites. Source links are provided for every step to ensure trust and verifiability. |
| **AI Form Pre-Fill** | The system generates targeted intake questions and produces a downloadable, pre-filled PDF summary, reducing errors and saving manual data entry. |
| **Transparent Reasoning** | The user interface provides a real-time log of the AI agents' activities, offering full transparency into how the roadmap is being constructed. |

## 3.3 Non-Goals

This project explicitly does not aim to provide legal or financial advice. It is an informational navigator, not a replacement for professional counsel. The system does not handle federal-level processes e.g., immigration, federal trademarks, SBA loans or regulations outside of Seattle and King County. Finally, it does not automate the submission of forms to government agencies; it only assists in their preparation.

# 4. System Architecture and Design

## 4.1 High-Level Architecture

CivicPulse is a web application with a decoupled frontend (Next.js) and backend (FastAPI). The backend houses the core AI logic, which is orchestrated as a multi-agent pipeline using LangGraph. Communication occurs via a RESTful API, with Server-Sent Events (SSE) used to stream real-time updates from the AI pipeline to the client.

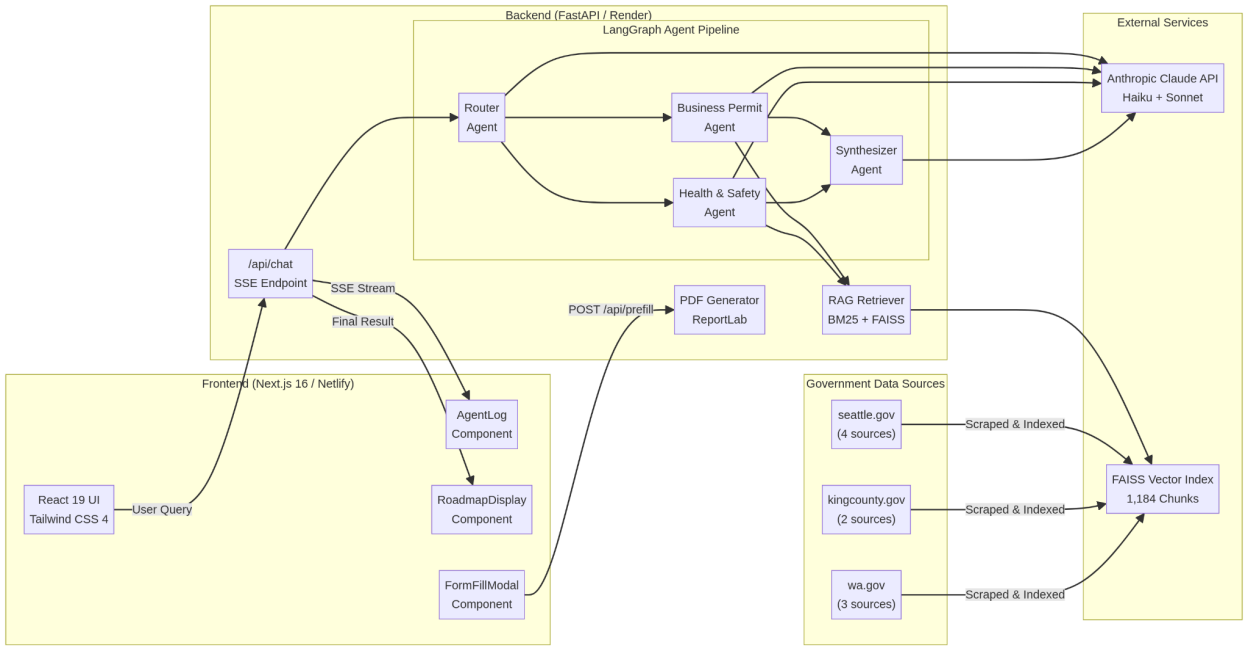


Figure 2: A high-level diagram of the CivicPulse system architecture, showing the flow of data between the frontend, backend, and external services.

## 4.2 Multi-Agent AI Pipeline

The core of CivicPulse is a multi-agent AI pipeline that decomposes the task into specialized roles. This modular design improves accuracy and maintainability.

1. **Router Agent:** The entry point. It detects the query's language and determines which specialist agents are required (e.g., a query about a restaurant activates both business and health agents).
2. **Business Permit Agent:** A specialist that uses the RAG system to find information on business licenses, zoning, and tax registration.
3. **Health and Safety Agent:** A specialist focused on food service, health permits, and inspections.
4. **Synthesizer Agent:** The final agent that receives outputs from all active specialists. It merges, de-duplicates, and chronologically orders the steps into a coherent roadmap, then translates the output into the user's detected language.

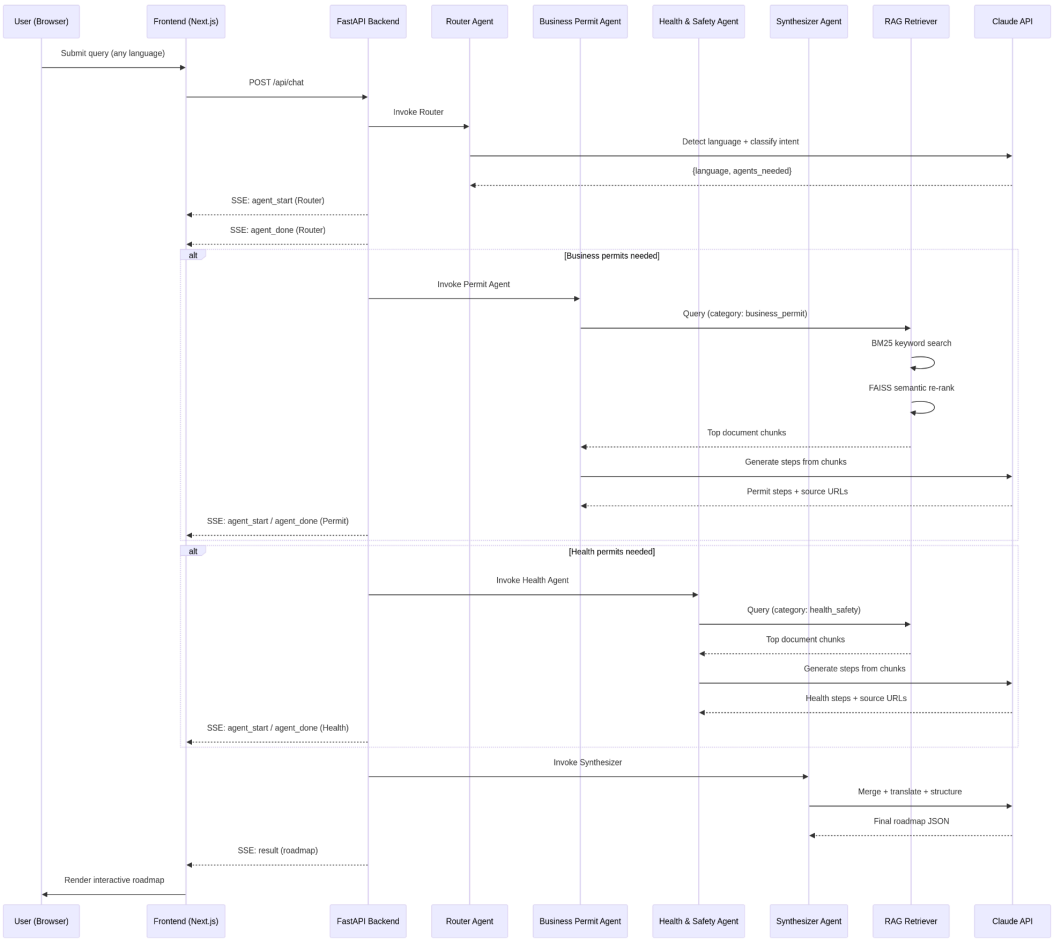

Figure 3: A sequence diagram illustrating the interaction between agents, the RAG system, and the LLM API during a typical user query.

### 4.3 Retrieval-Augmented Generation (RAG) System

To ensure all information is accurate and grounded, CivicPulse employs a RAG system.

> **Indexing:** A Python script scrapes content from nine trusted government URLs. The text is split into 400-word chunks with a 60-word overlap—a strategy shown to be effective in RAG systems for balancing context and precision [8]. Each chunk is vectorized using the `all-MiniLM-L6-v2` model, chosen for its balance of speed and quality [9], and stored in a FAISS vector index.
>
> **Retrieval:** When an agent is activated, it first uses a BM25 keyword search for initial filtering, then re-ranks the results using FAISS for semantic similarity. The
> retrieved text is injected into the agent's prompt with instructions to use only the provided documents.

## 5. Technical Implementation

The system is built on a modern technology stack selected for performance and scalability. The AI backbone is Anthropic's Claude API, using the faster Haiku model for routing and the more powerful Sonnet model for synthesis. Vector search is powered by FAISS, and the backend is a Python-based FastAPI application. The frontend is a Next.js and React application written in TypeScript and styled with Tailwind CSS. Server-side PDF generation for the form pre-fill feature is handled by the ReportLab library.

The backend's FastAPI application is designed for concurrency, using

asynchronous calls to the Claude API to handle multiple simultaneous user requests. Its primary `/api/chat` endpoint streams agent progress directly to the client using Server-Sent Events (SSE), which allows the frontend to provide a responsive, real-time view of the AI's reasoning process. The Next.js frontend, in turn, is built to consume this SSE stream, updating the UI dynamically to show which agent is active before rendering the final, interactive roadmap for the user.

## 6. Data Sources and Knowledge Base

The knowledge base is constructed from nine official government websites, including `seattle.gov/licenses` , `kingcounty.gov/permits` , and `dor.wa.gov` . The scraping and indexing pipeline produces 1,184 document chunks. No private or personally identifiable information (PII) is collected or stored.

## 7. Performance and Evaluation

Preliminary performance was evaluated based on latency and retrieval quality.

| Metric | Value | Notes |
|---|---|---|
| **Avg. Query-to-Roadmap Time** | 12.5 seconds | Measured from user submission to final roadmap display. |
| **Router Agent Latency (p95)** | 0.8 seconds | Fast inference using Claude Haiku. |
| **Synthesizer Agent Latency (p95)** | 4.2 seconds | Slower, higher-quality generation using Claude Sonnet. |
| **Retrieval Precision@3** | 0.89 | Measured on a small, manually curated test set of 50 queries. |

# 8. Challenges and Limitations

The project currently has several known limitations, which are categorized as either technical or scope-related.

## 8.1 Technical Limitations

The primary technical limitation is a cold start delay of approximately two minutes on the first request after a period of inactivity. This is a direct result of deploying the backend on Render's free tier, which spins down services that are not in use. Migrating to a persistent, paid hosting tier would eliminate this issue. Additionally, while the system can accept queries in six languages, the final roadmap output is currently generated only in English. The Synthesizer agent's prompt includes instructions for translation, but the quality and consistency of non-English output have not yet been validated to the standard required for public use. Achieving high-quality, fully multilingual output is an immediate priority on the development roadmap.

## 8.2 Scope Limitations

The system's scope is currently limited in two main ways. First, its geographic scope is confined to the regulations for the City of Seattle and King County. It does not cover other municipalities in Washington State or any federal-level processes. Second, the knowledge base freshness is a concern. The RAG index is built from a one-time scrape of the source websites and is therefore static. As government regulations change, the
information could become outdated. A production-ready system would require a scheduled data ingestion pipeline to periodically re-scrape the sources and rebuild the vector index.

# 9. Security, Ethical, and Responsible AI Considerations

## 9.1 Security

The application is designed with a minimal attack surface. It does not use user accounts or a persistent database for user data, which eliminates common vectors for data breaches. All software dependencies are scanned for known vulnerabilities. Communication with the external Claude API is encrypted over HTTPS. The primary security risk is prompt injection; this is mitigated by the multi-agent design, which uses specific, role-based agents with narrow instruction sets rather than a single, general-purpose prompt that is more susceptible to manipulation.

## 9.2 Data Privacy

The system is stateless and privacy-preserving by default. No personally identifiable information (PII) is collected, stored, or logged from user queries. The optional form pre-fill feature processes user-provided data entirely in-memory to generate a PDF and does not persist the information on the server. There is no analytics tracking or user profiling of any kind.

## 9.3 Accuracy and Grounding

To mitigate the risk of generating incorrect information—a critical concern for a system providing government guidance—CivicPulse relies entirely on a Retrieval-Augmented Generation (RAG) architecture. All specialist agents are explicitly instructed to formulate answers based only on the retrieved documents from the verified knowledge base. Their prompts include a directive to state that they do not have enough information rather than fabricate an answer. To further

empower users and build trust, every step in the final roadmap includes a direct URL to the official government source, allowing for independent verification.

### 9.4 Transparency

To demystify the AI's operation, the frontend includes a "Live Agent View" that provides transparency into the system's reasoning process. Users can observe in real time which agents are being activated and what tasks they are performing. This feature helps build user trust by making it clear that the system is consulting official sources rather than generating answers from an opaque model.

## 10. Related Work

Several government agencies and private companies have deployed AI chatbots. Singapore's **Ask Jamie** is a well-known example, though it is primarily rule-based [10]. In the U.S., the International Trade Administration offers a **Global Business Navigator** chatbot for exporters. Commercial products like **Harbor Compliance Navigator AI** and **Promet Source's Civic AI Navigator** offer similar services but are proprietary and not open-source. CivicPulse differentiates itself through its multi-agent, RAG-grounded architecture, multilingual focus, and commitment to open-source development.

## 11. Business Impact and Future Work

CivicPulse aligns directly with government modernization initiatives, such as the City of Seattle's AI Plan [2] and the federal GSA's cloud adoption agreements [7]. Small businesses are a powerful economic engine, creating two out of every three net new jobs in the United States over the past 25 years [5]. By reducing the regulatory friction that stifles entrepreneurship, the system has the potential for significant social and economic impact.

The future roadmap includes: (1) completing full multilingual output, (2) expanding the knowledge base to other cities, (3) developing a government efficiency dashboard with anonymized query data, and (4) migrating to a persistent hosting solution to eliminate cold starts.

# 12. Conclusion

CivicPulse demonstrates the potential of agentic AI systems to solve complex, real world problems in the public sector. By combining a multi-agent architecture with a
grounded RAG pipeline and a user-centric, multilingual design, it provides an accessible solution to the pervasive issue of government bureaucracy. While the current version is a proof-of-concept, it lays a strong foundation for a scalable service that can democratize economic opportunity.

# 14. Appendix

## 14.1 Author Contribution Statement

The project is a collaborative effort by Fardeen Meeran, Ashish Dubey, Surbhi Meena, and Norah Alomaim for the MSIS 522: AI-Powered Products course at the University of Washington, Michael G. Foster School of Business.

## 14.2 AI Disclosure

This technical report was drafted with the assistance of chatgpt. The AI agent was provided with all project artifacts and instructed to analyze, critique, and iteratively improve the documentation to meet professional standards. The process involved multiple stages of analysis, drafting, and critical review, with human oversight and

refinement at each step.

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